Billing app document

Billing Application – Functional & Calculation Logic Document

1. Overview

Application Name: Billing Application
Purpose:
To create, manage, and track customer invoices, calculate taxes and totals, record payments, and generate billing reports.

Target Users:

- Admin
- Billing Staff
- Accountant (read-only / reports)

Assumptions:

- Single currency (can be extended)
- Percentage-based tax (GST/VAT)
- Line-item–based invoices
- Decimal precision: 2 places (round half up)

2. User Roles & Permissions

Role	Permissions
Admin	Full access, configuration, reports
Billing Staff	Create/edit invoices, customers
Accountant	View invoices, reports only

3. Page-wise Functional Flow & Calculation Logic

Login Page

Purpose

Authenticate users.

Fields

- Username
- Password

Logic

- Validate credentials

No billing calculation on this page.

3. Dashboard Page

Purpose

Provide billing summary.

Displayed Metrics (Calculated)

Metric	Logic
Total Invoices (Today)	Count of invoices created today
Total Sales	Sum of Invoice Grand Total
Total Tax	Sum of Invoice Tax Amount
Pending Payments	Sum of unpaid invoice balances

3.1 Customer Management Page

Purpose

Create and manage customers.

Fields

- Customer Name
- Address
- Phone

- Email
- GST type
- GST
- state (with sate code)

Credit Balance

- opening balance (payable & receivable)

Additional Fields label *4 (to be print on invoice page)

No calculations performed.

3.4 Product (Item)/ Service Management Page

Purpose

Define billable items.

Fields

- Product Name
- HSN code
- category
- sale Price (sale price -purchase price - tax = profit(%))
- purchase price
- wholesale price & Qty
- Tax Rate (%)
- Unit of Measure

No calculations here; used later in invoices.

3.5 Invoice Creation Page (Core Billing Logic)

Purpose

Create customer invoices.

3.5.1 Invoice Header Fields

- Invoice Number (auto-generated)
- Invoice Date
- Customer name
- Due Date

3.5.2 Invoice Line Items Table

Each row contains:

Field	Description
Product	Selected product/service
Quantity	Numeric
Unit Price	From product (editable)
Discount (%)	Optional
Tax Rate (%)	From product
Round off	nearest value of 1
Line Total	Calculated

Line Item Calculation Logic

Step 1: Gross Amount

GrossAmount = Quantity × UnitPrice

Step 2: Discount Amount

DiscountAmount = GrossAmount × (Discount% / 100)

Step 3: Taxable Amount

TaxableAmount = GrossAmount - DiscountAmount

Step 4: Tax Amount

TaxAmount = TaxableAmount × (TaxRate% / 100)

Step 5: Line Total

LineTotal = TaxableAmount + TaxAmount

3.5.3 Invoice Summary Section

Field	Calculation
Subtotal	Σ TaxableAmount
Total Discount	Σ DiscountAmount
Total Tax	Σ TaxAmount
Shipping/Other Charges	Manual input
Grand Total	Final payable amount

Invoice Total Formula

GrandTotal = Subtotal + TotalTax + OtherCharges

Rounded to 2 decimal places.

3.6 Invoice View / Print Page

Purpose

Display final invoice.

Displays

- Invoice details
- Line items
- Subtotal
- Tax breakdown
- Grand Total
- Paid Amount
- Balance Due

Balance Due Calculation

BalanceDue = GrandTotal - PaidAmount

3.7 Payment Page

Purpose

Record customer payments.

Fields

- Invoice Number
- Payment Date
- Payment Mode (Cash, Card, Bank)
- Paid Amount

Payment Logic

Partial Payment Supported

RemainingBalance = PreviousBalance - PaidAmount

Invoice Status Logic

Condition	Status
PaidAmount = 0	Unpaid
PaidAmount < GrandTotal	Partially Paid
PaidAmount = GrandTotal	Paid

3.8 Credit Note / Refund Page

Purpose

Handle returns or billing errors.

Logic

RefundAmount ≤ Original Invoice Amount

AdjustedBalance = OriginalBalance - RefundAmount

3.9 Reports Page

Common Reports & Logic

Sales Report

TotalSales = Σ Invoice.GrandTotal (Date Range)

Tax Report

TotalTaxCollected = Σ Invoice.TotalTax

Outstanding Payments Report

OutstandingAmount = Σ BalanceDue where BalanceDue > 0

4. Sale

- Sale invoice
- Estimate / quotation
- Payment -in entry
- Sale order
- Delivery challan
- Sale return / Credit note(refer to page 3.8)

4. Purchase

- Purchase
- Payment - out entry
- Purchase order
- Purchase return /Debit note

5. Cash & bank

- Bank account (formal entry - calculation from (sale invoice , payment -in))(opening balance as capital investment)(digital amount)
- Cash in Hand (on table money)
- Cheque

6. Non-Functional Requirements

- Decimal precision: 2 digits
- Secure authentication
- Role-based access
- PDF invoice generation
- Audit trail for edits

7. Report

- Sale
- Purchase
- All fields report
- payment -in
- payment -out
- delivery challan
- credit note
- debit note

- sale order
- purchase order

Data fetch from all the required fields for specific

8. Future Enhancements

- Multi-currency support
- Recurring invoices
- Subscription billing
- Automatic tax rules
- API integration

Next step?

If you want, I can:

- Convert this into **SRS format**
- Create **database schema**
- Write **API logic**
- Provide **sample UI wireframes**
- Customize for **GST / VAT / country-specific tax rules**
- **job role based system according to the price management system**